2 **(a)** Explain what is meant by the following terms when used in the context of forces.

(i) Centre of gravity

.....

.....

..... [1]

(ii) Friction

.....

.....

..... [1]

(b) Fig. 2.1 shows a man performing a bungee jump.

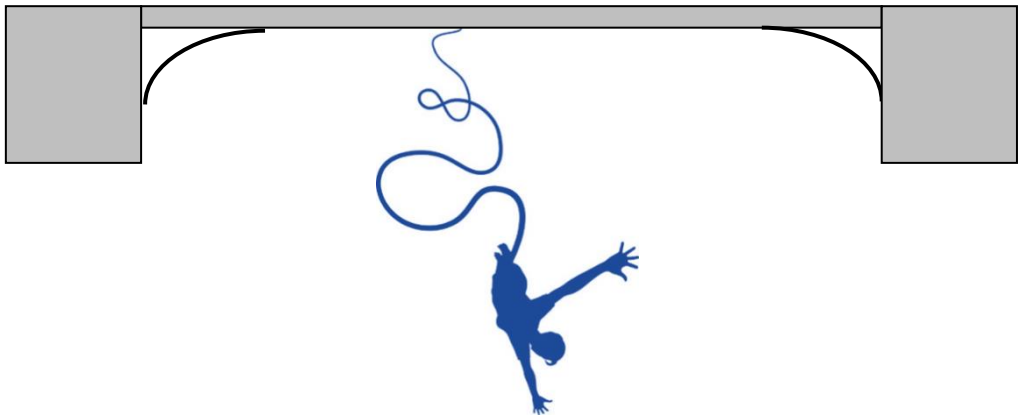

Fig. 2.1

He falls 42 m before the elastic rope secured to him starts to exert some force on him. The force-extension graph for this rope is shown in Fig. 2.2.

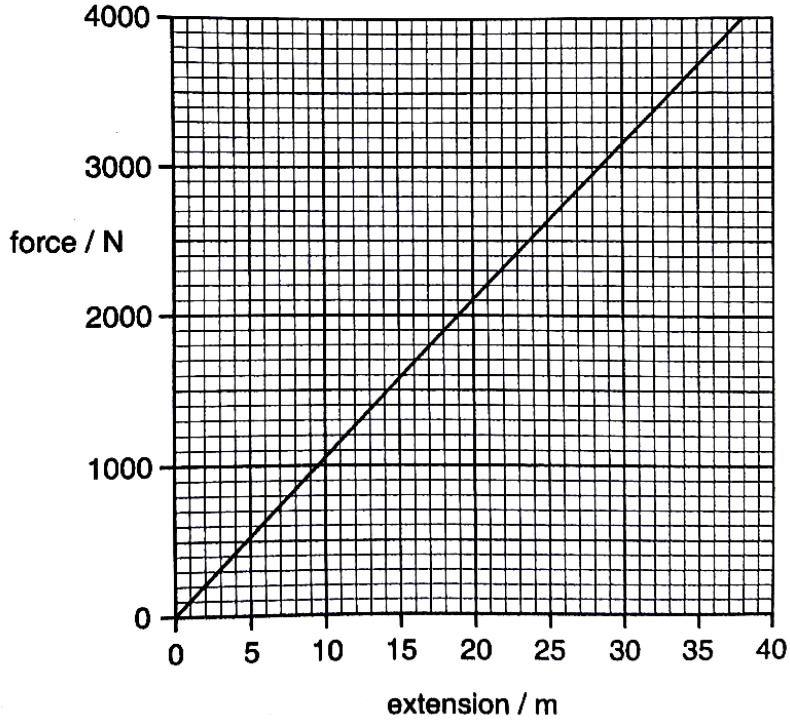

Fig. 2.2

The total distance the man falls before he stops for the first time is 78 m.

(i) Deduce

1. the extension of the rope when the man stops for the first time, and

extension = …….. m [1]

2. the elastic potential energy stored in the rope at this time.

elastic potential energy = …….. J [2]

(ii) Show that the man has a mass of approximately 90 kg.

[2]